

week06报告

所有相关文件可以在 [这里](#) 找到。

所有图片中的内容，包括代码、日志、控制台输出，均可以在该文档中找到。

带Pro的是拓展任务。

一、异常类文件

1. InvalidAgeException.java

- 类型：Checked Exception
- 功能：
 - 表示年龄验证失败的异常
 - 当检测到年龄 < 18 时抛出
 - 强制调用方必须处理该异常

```
public class InvalidAgeException extends Exception {  
    public InvalidAgeException(String message) {  
        super(message);  
    }  
}
```

2. InvalidNameException.java

- 类型：Unchecked Exception
- 功能：
 - 表示姓名验证失败的异常
 - 处理以下两种情况：
 - 姓名为空或空字符串
 - 姓名长度 < 3 个字符
 - 可自动传播调用栈

```
public class InvalidNameException extends RuntimeException {  
    public InvalidNameException(String message) {  
        super(message);  
    }  
}
```

3. ApplicationException.java（拓展任务）

- 类型：Checked Exception
- 功能：
 - 高层业务异常包装器
 - 用于异常转译模式

- 包含原始异常信息（通过 cause 参数）
- 统一系统异常出口

```
public class ApplicationException extends Exception {  
    public ApplicationException(String message, Throwable cause) {  
        super(message, cause);  
    }  
}
```

二、核心业务类

UserRegistration.java

```
public class UserRegistration {  
  
    public void validateAge(int age) throws InvalidAgeException {  
        if (age < 18) {  
            throw new InvalidAgeException("Illegal age!");  
        }  
    }  
  
    public void validateName(String name) {  
        if (name == null || name.trim().length() < 3) {  
            throw new InvalidNameException("Illegal name!");  
        }  
    }  
  
    public void registerUser(String name, int age) throws InvalidAgeException {  
        System.out.println("Attempting to register user: Name='" + name + "', Age=" + age);  
  
        validateName(name);  
        validateAge(age);  
  
        System.out.println("Registration successful!");  
        System.out.println("User Details: Name = " + name + ", Age = " + age);  
    }  
}
```

UserRegistrationPro.java（扩展任务）

```
import java.util.logging.Logger;  
  
public class UserRegistrationPro {  
    private static final Logger logger =  
        Logger.getLogger(UserRegistrationPro.class.getName());  
  
    public void validateAge(int age) throws InvalidAgeException {  
        if (age < 18) {  
            logger.warning("Age illegal: " + age);  
            throw new InvalidAgeException("Illegal age!");  
        }  
    }  
}
```

```

    }
}

public void validateName(String name) {
    if (name == null || name.trim().length() < 3) {
        logger.warning("Name too short: " + name);
        throw new InvalidNameException("Short name!");
    }
    validateFormat(name);
}

// 更深层次的验证方法
private void validateFormat(String name) throws InvalidNameException {
    if (!name.matches("[\\u4E00-\\u9FA5A-Za-z]+")) {
        throw new InvalidNameException("Name illegal!");
    }
}

public void registerUser(String name, int age) throws ApplicationException {
    try {
        validateName(name);
        validateAge(age);
        logger.info("Registration successful!" + name);
        logger.info("User Details: Name = " + name + ", Age = " + age);
    } catch (InvalidAgeException | InvalidNameException e) {
        throw new ApplicationException("Registration fail!", e);
    }
}
}

```

三、系统入口类

Main.java

```

public class Main {

    public static void main(String[] args) {

        startRegistration( "fQwQf", 18);

        startRegistration( "fQ", 18);

        startRegistration( "fQwQf", 17);

    }

    public static void startRegistration(String name, int age) {
        UserRegistration userRegistration = new UserRegistration();
        try {
            userRegistration.registerUser(name, age);
        } catch (InvalidAgeException e) {

```

```

        System.err.println("Registration Failed: " + e.getMessage());
    } catch (InvalidNameException e) {
        System.err.println("Registration Failed: " + e.getMessage());
    }
}
}
}

```

MainPro.java (扩展任务)

```

import java.util.logging.*;
import java.io.IOException;

public class MainPro {
    private static final Logger logger = Logger.getLogger(MainPro.class.getName());

    static {
        FileHandler fileHandler;
        try {
            fileHandler = new FileHandler("app.log");
        } catch (IOException e) {
            throw new RuntimeException(e);
        }
        fileHandler.setFormatter(new SimpleFormatter());
        logger.addHandler(fileHandler);
        logger.setUseParentHandlers(false); // 关闭控制台输出
    }

    // 多层调用链
    public static void serviceLayer(String name, int age) throws ApplicationException {
        try {
            businessLogicLayer(name, age);
        } catch (ApplicationException e) {
            logger.log(Level.SEVERE, "Service layer error!", e);
            throw new ApplicationException("Service error!", e);
        }
    }

    private static void businessLogicLayer(String name, int age) throws ApplicationException {
        UserRegistrationPro registration = new UserRegistrationPro();
        registration.registerUser(name, age);
    }

    public static void main(String[] args) {
        test("fQwQf", 18); // 正常注册
        test("fQwQf6", 18); // 非法字符
        test("fQ", 16); // 姓名过短+年龄不足
    }

    private static void test(String name, int age) {
        try {

```

```

        serviceLayer(name, age);
        System.out.println("Success: " + name );
    } catch (ApplicationException e) {
        System.out.println("Fail: " + name + ": " + e.getMessage());
    }
}
}
}

```

四、运行结果

普通任务

```

/home/fqwqf/.jdk/openjdk-24/bin/java -javaagent:/home/fqwqf/桌面/idea-IC-
243.25659.59/lib/idea_rt.jar=36961 -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -
Dsun.stderr.encoding=UTF-8 -classpath /home/fqwqf/桌
面/project/WHU_OOP_2025/week06/out/production/week06 Main
Attempting to register user: Name='fQwQf', Age=18
Registration successful!
User Details: Name = fQwQf, Age = 18
Attempting to register user: Name='fQ', Age=18
Attempting to register user: Name='fQwQf', Age=17
Registration Failed: Illegal name!
Registration Failed: Illegal age!

```

进程已结束，退出代码为 0

扩展任务

```

/home/fqwqf/.jdk/openjdk-24/bin/java -javaagent:/home/fqwqf/桌面/idea-IC-
243.25659.59/lib/idea_rt.jar=45557 -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -
Dsun.stderr.encoding=UTF-8 -classpath /home/fqwqf/桌
面/project/WHU_OOP_2025/week06/out/production/week06 MainPro
4月 01, 2025 11:29:29 下午 UserRegistrationPro registerUser
信息: Registration successful!fQwQf
4月 01, 2025 11:29:29 下午 UserRegistrationPro registerUser
信息: User Details: Name = fQwQf, Age = 18
4月 01, 2025 11:29:29 下午 UserRegistrationPro validateName
警告: Name too short: fQ
Success: fQwQf
Fail: fQwQf6: Service error!
Fail: fQ: Service error!

```

进程已结束，退出代码为 0

五、生成文件

app.log（扩展任务）

- 生成方式：通过 FileHandler 自动创建
- 内容结构：

```
4月 01, 2025 10:16:05 下午 MainPro serviceLayer
严重: Service layer error!
ApplicationException: Registration fail!
    at UserRegistrationPro.registerUser(UserRegistrationPro.java:35)
    at MainPro.businessLogicLayer(MainPro.java:32)
    at MainPro.serviceLayer(MainPro.java:23)
    at MainPro.test(MainPro.java:43)
    at MainPro.main(MainPro.java:37)
Caused by: InvalidNameException: Name illegal!
    at UserRegistrationPro.validateFormat(UserRegistrationPro.java:24)
    at UserRegistrationPro.validateName(UserRegistrationPro.java:18)
    at UserRegistrationPro.registerUser(UserRegistrationPro.java:30)
    ... 4 more

4月 01, 2025 10:16:05 下午 MainPro serviceLayer
严重: Service layer error!
ApplicationException: Registration fail!
    at UserRegistrationPro.registerUser(UserRegistrationPro.java:35)
    at MainPro.businessLogicLayer(MainPro.java:32)
    at MainPro.serviceLayer(MainPro.java:23)
    at MainPro.test(MainPro.java:43)
    at MainPro.main(MainPro.java:38)
Caused by: InvalidNameException: Short name!
    at UserRegistrationPro.validateName(UserRegistrationPro.java:16)
    at UserRegistrationPro.registerUser(UserRegistrationPro.java:30)
    ... 4 more
```

六、总结

这次作业让我真正理解了异常处理的意义。从最初机械地写try-catch，到后来能区分何时该用受检异常（如年龄校验必须处理）、何时该用运行时异常（如姓名格式问题），这种思维转变就像突然读懂了Java的设计范式。

在搭多层调用链时，异常像接力棒一样在各层传递，特别是用ApplicationException包装原始异常时，既保持了上层调用的简洁，又通过日志文件保留了完整的错误线索，这种设计比直接打印错误信息优雅得多。

这次实践让我意识到，好的异常处理就像给程序系上安全带——平时感觉不到存在，但遇到意外时能稳稳兜住问题。虽然现在的实现还比较基础，但已经为后续学习更复杂的错误处理机制打下了扎实的基础。